

A Sync-Point MDP and Screened Rollout for Truck-Walk Delivery Under Uncertainty

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Abstract—This paper studies execution-phase decision making for truck-walk last-mile delivery under uncertainty. The benchmark instances come from the SP_{*} urban-rural assisted-delivery set, where truck driving can be slower than walking on some edges. Rather than considering only a deterministic route-planning model, the paper formulates execution as a finite-horizon Markov decision process defined at synchronization points where the truck and worker are colocated. The state tracks the current node, the remaining customers, the traffic regime, and elapsed time. Actions correspond to coordinated line and loop truck-walk sorties. Truck travel-time uncertainty is modeled through a regime-aware heteroskedastic stochastic model built from the benchmark drive-walk structure, and customer service times are modeled as stochastic stop-level delays. The solution method is a screened staged rollout policy that combines an offline sortie library, online action screening, deterministic lookahead, and Monte Carlo action evaluation with common random numbers. Across nine induced-instance main settings, rollout reduced mean completion time by an average of 2.86% relative to deterministic mean replanning. On three selected 50-customer instances, rollout achieved an average 2.89% improvement relative to mean replanning while remaining computationally feasible. Robustness analyses show that the magnitude of the rollout advantage depends on candidate-action richness and on how uncertainty enters the execution process. Overall, the results show that a sync-point MDP with screened rollout can extend deterministic truck-walk planning to an online setting with uncertainty.

I. INTRODUCTION

Autonomous-vehicle-assisted last-mile delivery models typically assume that travel times are known in advance and optimize a complete route before execution. That assumption is useful for planning, but it does not capture the online coordination challenge that arises when realized truck travel times and stop service times differ from their nominal values. In a truck-walk system, these execution deviations matter because the truck and worker repeatedly separate and reunite. A poor local coordination decision can create substantial waiting and route delay.

This paper extends the deterministic truck-walk setting to an online decision problem. The central question is whether an online policy based on a finite-horizon Markov decision process can outperform a fixed deterministic plan or deterministic replanning based only on mean times. The paper uses the SP_{*} benchmark family because these instances explicitly contain settings where truck driving may be slower than walking, so synchronization decisions are operationally important.

The main contribution lies in problem formulation and algorithm design. The study introduces a sync-point MDP with explicit state, action, transition, and reward definitions. It also develops a benchmark-informed uncertainty model tied to the SP_{*} drive-walk geometry and a scalable screened rollout method that remains computationally viable on selected 50-customer instances. The contribution is not a new theoretical class of rollout algorithms. Instead, the paper adapts rollout to the truck-walk setting and evaluates its value in a structured computational study.

II. BACKGROUND AND RELATED WORK

This paper lies at the intersection of assisted-delivery routing and approximate dynamic programming for stochastic routing.

On the assisted-delivery side, Reed et al. [1] and Reed et al. [2] study autonomous-vehicle-assisted last-mile delivery in urban and urban-to-rural settings, showing both the promise of the assisted-delivery concept and the importance of geographic structure. Zhang [3] studies the capacitated autonomous-vehicle-assisted delivery problem using branch-and-price and provides the deterministic structural background for truck-worker coordination, feasible sortie patterns, and benchmark instances.

On the dynamic-routing side, Secomandi [4] introduced rollout ideas for stochastic vehicle routing. Goodson et al. [5] developed multiple rollout variants for dynamic multivehicle routing with stochastic demand and duration limits, including pre-decision and post-decision viewpoints. Goodson et al. [6] further studied rollout design choices and heuristic components. Ulmer et al. [7] showed how offline-online approximate dynamic programming can improve dynamic vehicle routing under uncertainty. Zhang et al. [8] also provide a relevant example of a routing MDP solved by rollout in a stochastic environment.

These studies motivate both the formulation and the computational study in this paper. Existing assisted-delivery work is primarily deterministic and focuses on route design before execution. By contrast, the rollout and approximate dynamic programming literature studies online decision making under uncertainty, but not the truck-walk synchronization structure considered here. This paper connects these two lines of work by treating truck-walk delivery as an execution-phase decision problem and by evaluating an online recourse policy on benchmark instances from the assisted-delivery literature.

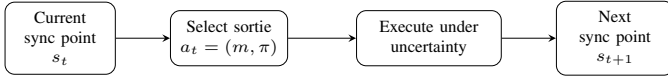


Fig. 1. Sync-point decision cycle. A decision is made only when the truck and worker are colocated.

The same literature also motivates comparisons against static and replanning baselines, sensitivity analysis over uncertainty levels, a study of runtime versus rollout budget, and scalability tests on larger instances.

III. PROBLEM FORMULATION

A. System Description

The system contains one truck, one walking delivery worker, one depot, and a set of customers. The truck can transport the worker to a customer, after which the worker serves one or more customers on foot and later reunites with the truck. The objective is to minimize expected total completion time.

To keep the decision process tractable, decisions are made only at *synchronization points* where the truck and worker are together. This turns the execution problem into a finite sequence of local coordination decisions. Since every action serves at least one customer, the horizon is finite and is bounded above by the number of remaining customers. Figure 1 illustrates this decision cycle.

B. MDP Formulation

The problem is modeled as a finite-horizon MDP (S, A, T, R) .

1) *State Space*: A state is

$$s = (i, U, g, \tau), \quad (1)$$

where i is the current synchronization node, U is the set of unserved customers, $g \in \{\text{low}, \text{normal}, \text{high}\}$ is the current traffic regime, and τ is the elapsed route time. The initial state is $(0, C, \text{normal}, 0)$, where node 0 is the depot and C is the full customer set.

2) *Action Space*: An action is a coordinated truck-walk sortie

$$a = (m, \pi), \quad (2)$$

where m is the action mode and $\pi = (k_1, \dots, k_r)$ is the worker walking sequence. In a line action, the truck drops the worker at k_1 , the worker serves path π , and the truck reunites with the worker at k_r . In a loop action, the truck drops the worker at k_1 , waits there, and the worker serves π and returns to k_1 . The sortie size is bounded by a parameter q , so $r \leq q$.

The feasible action set depends on the current state. For state $s = (i, U, g, \tau)$,

$$\begin{aligned} A(s) = \{ (m, \pi) : m \in \{\text{line}, \text{loop}\}, \pi = (k_1, \dots, k_r), \\ 1 \leq r \leq q, \{k_1, \dots, k_r\} \subseteq U, \\ k_p \neq k_{p'} \text{ for } p \neq p' \}. \end{aligned} \quad (3)$$

Let $B(a) = \{k_1, \dots, k_r\}$ denote the set of customers served by action a . Let $j(a) = k_r$ for a line action and $j(a) = k_1$ for a loop action.

3) *Transition Function*: Let the current state be $s = (i, U, g, \tau)$ and let action a be executed. The truck nominal edge means come from the SP_* driving matrix, but the actual truck time depends on the traffic regime and on an edge friction index derived from the drive-walk ratio. Service times are stochastic, while walking times remain deterministic.

For a walking sequence $\pi = (k_1, \dots, k_r)$, define

$$W_{\text{line}}(\pi) = \sum_{p=1}^{r-1} w_{k_p, k_{p+1}} \quad (4)$$

and

$$W_{\text{loop}}(\pi) = W_{\text{line}}(\pi) + w_{k_r, k_1}, \quad (5)$$

where w_{uv} is the deterministic walking time from customer u to customer v .

For a line action, the one-step cost is

$$c(s, a, \xi) = D_{i, k_1}(g, \xi) + f + \max\{D_{k_1, k_r}(g, \xi), W_{\text{line}}(\pi) + S(\pi, \xi)\}, \quad (6)$$

where f is the loading or setup time, $D_{uv}(g, \xi)$ is the realized truck travel time from node u to node v under regime g , and $S(\pi, \xi)$ is the random service time accumulated along the walking sequence.

For a loop action, the truck waits at the drop-off node, so the one-step cost becomes

$$c(s, a, \xi) = D_{i, k_1}(g, \xi) + f + W_{\text{loop}}(\pi) + S(\pi, \xi). \quad (7)$$

The next state is

$$s' = (j(a), U \setminus B(a), g', \tau + c(s, a, \xi)), \quad (8)$$

where g' is the next traffic regime sampled from a three-state Markov chain.

4) *Reward Function*: The reward is the negative one-step completion time:

$$R(s, a, s') = -c(s, a, \xi). \quad (9)$$

Equivalently, the objective is to minimize expected cumulative completion time.

Let $J(s)$ denote the optimal cost-to-go from state s . Then the finite-horizon decision problem can be written as

$$J(s) = \min_{a \in A(s)} \mathbb{E}[c(s, a, \xi) + J(s')]. \quad (10)$$

If all customers have been served, the process terminates after the truck returns to the depot. For a terminal state $(i, \emptyset, g, \tau)$, the terminal cost is the expected truck return time from node i to the depot under regime g .

IV. SOLUTION APPROACH

A. Instance Construction

The original SP_* instances contain one depot and 50 customers. The computational study uses these instances in two ways. Induced subinstances of size 10 to 20 customers are used for controlled experiments, and selected full-size 50-customer instances are used for scalability analysis. Each induced instance is created by uniform sampling without

replacement from the parent 50-customer instance. The main experiment matrix uses a fixed subinstance seed, while the seed-robustness study later varies this choice.

B. Uncertainty Model

The benchmark data are deterministic, so the paper constructs a stochastic execution layer on top of the nominal matrices rather than claiming to estimate empirical distributions from historical data. The $SP_{\star}$ driving matrix provides nominal truck means, and the $SP_{\star}$ walking matrix provides deterministic walking times.

Each state includes a traffic regime $g \in \{\text{low, normal, high}\}$, and the regime evolves according to a simple Markov chain. For customer-customer edges with positive walking times, the model uses the drive-walk friction ratio

$$\rho_{uv} = \frac{\bar{d}_{uv}}{\bar{w}_{uv}}, \quad (11)$$

where $\bar{d}_{uv}$ and $\bar{w}_{uv}$ are the nominal driving and walking times. Realized truck travel times are sampled from lognormal distributions whose means are scaled by the current regime and whose coefficients of variation increase on edges with larger normalized friction ratios. This links truck-side uncertainty to the benchmark structure rather than treating all edges as equally variable.

Service times are also sampled from a positive stop-level distribution, again using a lognormal model, with low, medium, and high variability settings. Walking times remain deterministic. This construction is synthetic, but it is benchmark-informed: the benchmark identifies where truck movement is structurally disadvantaged, and the stochastic layer amplifies that disadvantage during execution.

C. Candidate Action Generation

The full action space is combinatorial and cannot be enumerated online at large scale. To address this, the algorithm first constructs an offline sortie library. For each possible worker drop-off node, short promising line and loop paths are precomputed using nearest-neighbor expansion and beam search. At each decision point, this library is filtered by the remaining-customer set and the current truck node, and only the top-ranked actions are retained. This is the main mechanism that allows the method to handle 50-customer instances heuristically.

D. Baseline Policies

Open-loop mean-time plan: This baseline computes a deterministic plan once at the start using nominal means and executes that plan without replanning.

Mean-time replanning: This baseline recomputes the next action at every synchronization point using a deterministic shallow lookahead. It uses the current state, remaining customers, and traffic regime, but still evaluates actions with mean quantities only.

Algorithm 1 Screened Staged Rollout at sync-point state s

Require: Current state s , offline sortie library $\mathcal{L}$, deterministic base policy π , rollout budgets $N_{\text{pilot}}, N_{\text{full}}$

Ensure: Next coordination action a^*

- 1: Generate feasible candidate actions $A(s)$ by filtering $\mathcal{L}$ using the remaining-customer set and current truck node
 - 2: Rank $A(s)$ by deterministic base-policy score
 - 3: Keep the top K screened actions
 - 4: Construct a state-dependent seed set so all candidate actions use common random numbers
 - 5: **for** each screened action $a \in A(s)$ **do**
 - 6: Estimate a pilot value $\hat{Q}_{\text{pilot}}(s, a)$ using N_{pilot} Monte Carlo samples
 - 7: **end for**
 - 8: Keep the top H survivor actions under $\hat{Q}_{\text{pilot}}$
 - 9: **for** each survivor action a **do**
 - 10: Estimate a refined value $\hat{Q}(s, a) = \frac{1}{N_{\text{full}}} \sum_{n=1}^{N_{\text{full}}} (c(s, a, \xi_n) + J^{\pi}(s'_n))$
 - 11: **end for**
 - 12: Return $a^* = \arg \min_a \hat{Q}(s, a)$
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E. Main Policy: Screened Staged Rollout

The main algorithm is a screened staged rollout policy. At each decision point, the method generates a screened candidate action set, ranks those actions with the deterministic base policy, evaluates the leading actions with a small pilot rollout budget, and retains only a small number of survivors for a fuller rollout evaluation. The chosen action is the one with the smallest estimated cost-to-go. The implementation also uses deterministic state-based random seeds so that candidate actions are compared under common random numbers, cached action decisions by state, and a stronger deterministic base policy than a purely myopic heuristic. This design follows the spirit of the rollout literature, which seeks a balance between policy quality and computational tractability.

V. EXPERIMENTAL DESIGN

The experiment matrix follows common patterns in stochastic routing and rollout papers.

A. Core Policy Comparison

The main comparison uses three policies: `open_loop_mean_plan`, `mean_replan`, and `rollout`. The induced-instance study uses three representative instance families, `SP_Cook_1`, `SP_Winnebago_1`, and `SP_Johnson_8`, at customer counts $n \in \{10, 15, 20\}$.

B. Uncertainty Sensitivity

Three uncertainty levels are tested, namely low, medium, and high. These levels vary the truck travel-time coefficient of variation, the service-time coefficient of variation, and the regime multipliers in the stochastic simulator.

TABLE I
MAIN COMPARISON ON THE LARGEST INDUCED INSTANCES ($n = 20$).
ENTRIES ARE MEAN TOTAL COMPLETION TIMES WITH 95% CONFIDENCE
INTERVALS.

Instance	Open	Replan	Rollout	Imp. vs Replan
SP_Cook_1	87.19 $\pm$ 2.19	87.19 $\pm$ 2.19	84.15 $\pm$ 3.74	3.49%
SP_Winnebago_1	92.93 $\pm$ 2.58	91.34 $\pm$ 2.22	89.34 $\pm$ 4.15	2.19%
SP_Johnson_8	213.72 $\pm$ 13.94	213.94 $\pm$ 13.42	216.94 $\pm$ 15.28	-1.40%

C. Runtime and Rollout-Budget Tradeoff

The rollout budget is varied across 4, 8, and 16 rollouts to study the marginal value of deeper lookahead versus computational effort.

D. Scalability

The main statistical experiments are run on induced instances of 10, 15, and 20 customers. Selected full-size 50-customer SP_* instances are then used to demonstrate scalability.

E. Metrics

The primary reported metric is mean total completion time. For the main, sensitivity, and scalability experiments, the paper also reports 95% confidence intervals. Secondary metrics include mean truck waiting time, mean worker waiting time, mean number of synchronization decisions, and mean online decision time. To keep the robustness tables compact, only mean rollout cost, improvement relative to mean replanning, and mean decision time are tabulated there. All three policies are evaluated with the same episode seeds within each experimental setting so that comparisons are made on a common set of stochastic scenarios.

VI. RESULTS

The core experiment matrix used 10 Monte Carlo episodes per setting on three representative instance families, namely SP_Cook_1, SP_Winnebago_1, and SP_Johnson_8. The main comparison used induced instance sizes $n \in \{10, 15, 20\}$. The sensitivity study used three uncertainty levels, and the budget study used three rollout budgets. Selected 50-customer instances were used for scalability. This core suite produced 900 episode-level observations and 90 aggregated policy summaries. The complete core suite finished in about 365.31 seconds on the test platform.

A. Main Policy Comparison

Across the nine main-comparison settings, rollout improved mean completion time over mean replanning by an average of 2.86%. The effect was strongest on the urban and mixed instances and weaker on the rural instance.

Figure 2 and Table I show that the rollout policy consistently improved upon mean replanning on the urban SP_Cook_1 instance, with gains of 5.58%, 4.58%, and 3.49% at $n = 10$, 15, and 20, respectively. The mixed SP_Winnebago_1 instance also benefited from rollout, especially at $n = 15$, where the gain over mean replanning reached 8.29%. The rural SP_Johnson_8 instance proved more challenging. Rollout

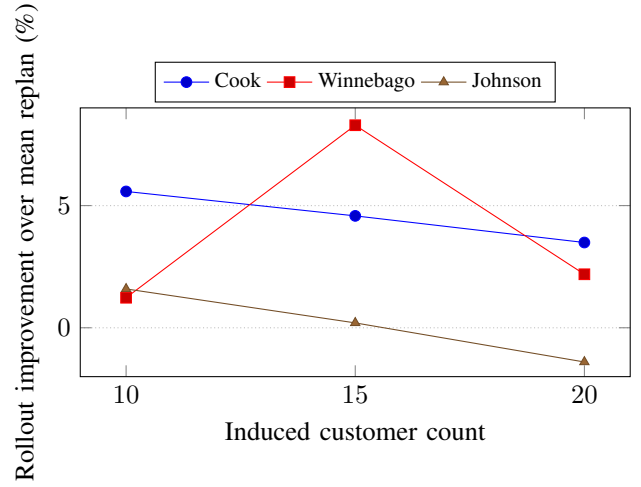


Fig. 2. Rollout improvement relative to mean replanning across induced instance sizes. Positive values favor rollout.

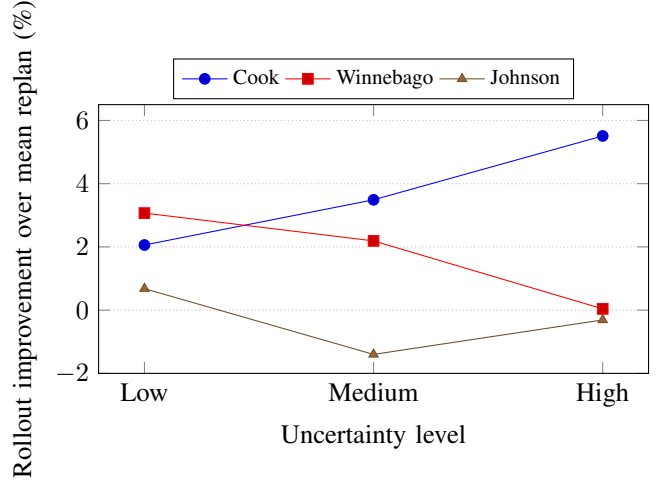


Fig. 3. Effect of uncertainty level on rollout improvement relative to mean replanning for induced instances with $n = 20$. Positive values favor rollout.

improved over the baselines at $n = 10$, was nearly tied at $n = 15$, and underperformed slightly at $n = 20$. This negative result is informative because it indicates that the current action-screening and base-policy design is not uniformly strong across geographic regimes.

B. Uncertainty Sensitivity

The uncertainty-sensitivity study used the $n = 20$ induced instances and low, medium, and high uncertainty settings.

Table II and Fig. 3 show that, on SP_Cook_1, rollout improvement over mean replanning increased from 2.06% under low uncertainty to 3.49% under medium uncertainty and 5.51% under high uncertainty. This is the clearest case where stochastic lookahead becomes more valuable as the environment becomes less predictable.

On SP_Winnebago_1, rollout remained better than the deterministic baselines under all three uncertainty levels, but

TABLE II

UNCERTAINTY SENSITIVITY ON INDUCED INSTANCES WITH $n = 20$. THE ROLLOUT COLUMN REPORTS MEAN COMPLETION TIME WITH A 95% CONFIDENCE INTERVAL.

Instance	Level	Rollout	Imp. vs Replan
SP_Cook_1	Low	83.26 ± 1.30	2.06%
SP_Cook_1	Medium	84.15 ± 3.74	3.49%
SP_Cook_1	High	85.30 ± 6.35	5.51%
SP_Winnebago_1	Low	86.15 ± 1.94	3.07%
SP_Winnebago_1	Medium	89.34 ± 4.15	2.19%
SP_Winnebago_1	High	94.51 ± 4.99	0.04%
SP_Johnson_8	Low	204.33 ± 5.59	0.68%
SP_Johnson_8	Medium	216.94 ± 15.28	-1.40%
SP_Johnson_8	High	233.70 ± 31.50	-0.31%

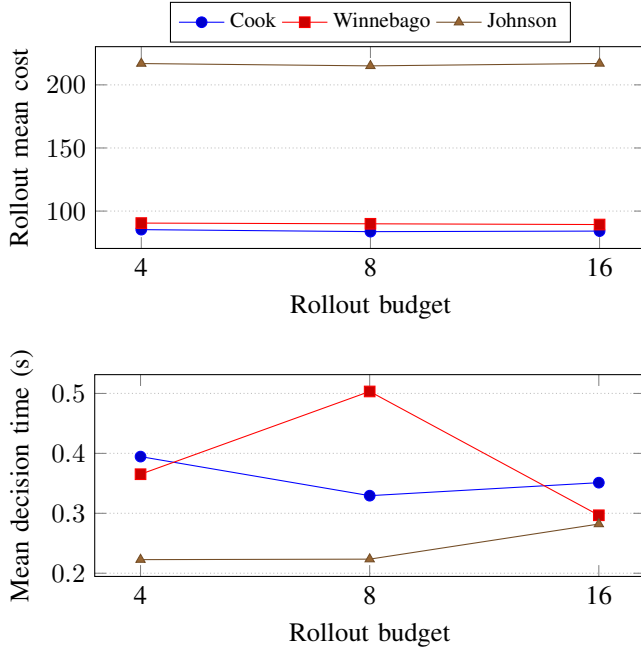


Fig. 4. Rollout budget tradeoff on induced instances with $n = 20$. The upper panel reports mean completion time, and the lower panel reports mean online decision time.

the improvement was more modest and not monotone. On SP_Johnson_8, rollout improved slightly under low uncertainty, but lost ground under medium and high uncertainty. Overall, the uncertainty results support the value of stochastic lookahead on some instance families, while also showing that the strength of that advantage is not uniform across geographies in the current implementation.

C. Rollout Budget and Runtime

The rollout-budget experiment compared 4, 8, and 16 rollouts on the $n = 20$ induced instances.

Table III and Fig. 4 indicate that moderate budgets are sufficient. On SP_Cook_1, 8 rollouts gave the best mean result, slightly outperforming both 4 and 16. On SP_Winnebago_1, 16 rollouts gave the best mean result, but the difference relative to 8 rollouts was small. On

TABLE III

ROLLOUT BUDGET TRADEOFF ON INDUCED INSTANCES WITH $n = 20$.

Instance	Budget	Rollout Mean	Decision Time (s)
SP_Cook_1	4	85.303	0.3946
SP_Cook_1	8	83.708	0.3294
SP_Cook_1	16	84.145	0.3511
SP_Winnebago_1	4	90.490	0.3652
SP_Winnebago_1	8	89.931	0.5033
SP_Winnebago_1	16	89.338	0.2967
SP_Johnson_8	4	216.885	0.2227
SP_Johnson_8	8	215.043	0.2235
SP_Johnson_8	16	216.944	0.2821

TABLE IV

SELECTED 50-CUSTOMER SCALABILITY RESULTS. ENTRIES ARE MEAN TOTAL COMPLETION TIMES WITH 95% CONFIDENCE INTERVALS.

Instance	Open	Replan	Rollout	Imp. vs Replan	Time (s)
SP_Cook_1	202.39 ± 2.45	204.83 ± 2.32	198.42 ± 3.75	3.13%	16.67
SP_Winnebago_1	191.90 ± 3.26	191.36 ± 3.50	185.79 ± 3.74	2.91%	6.25
SP_Johnson_8	330.15 ± 8.46	329.11 ± 12.93	320.47 ± 10.87	2.62%	5.57

SP_Johnson_8, none of the tested rollout budgets resolved the robustness issue.

Mean rollout decision times on induced instances remained modest, ranging from about 0.22 seconds to 0.50 seconds. This is slower than deterministic replanning, but it remains small on the scale of the present computational study.

D. Full-Size 50-Customer Scalability

The selected 50-customer runs provide the strongest evidence that the algorithm is not limited to very small instances. Using the same framework with a capped rollout budget of 8, the heuristic completed all three representative full-size instances successfully.

Table IV shows that the 50-customer results are notable for two reasons. First, rollout improved over both deterministic baselines on all three selected full-size instances, including the rural instance for which the induced-instance results were less favorable. Second, the policy remained computationally feasible. The online decision times are nontrivial, but they remain within a practical range for the present study. These scalability results are based on selected full-size instances and a capped rollout budget of 8. They should therefore be interpreted as evidence of computational viability rather than as an exhaustive benchmark study.

E. Discussion

Taken together, the results show that the sync-point MDP is a useful operational model for online recourse in assisted delivery. The screened rollout method is especially effective on the urban and mixed instances, where it consistently improves mean completion time. The rural instances remain the most challenging cases. Even so, the method performs well on the selected full-size rural instance. The budget study also suggests that moderate rollout budgets are sufficient in most settings, which is important for scalability.

TABLE V
ACTION-SCREENING ROBUSTNESS ON INDUCED 20-CUSTOMER
INSTANCES.

Instance	Screening	Rollout	Imp. vs Replan	Time (s)
SP_Cook_1	Aggressive	85.119	1.41%	0.079
SP_Cook_1	Baseline	83.057	3.80%	0.445
SP_Cook_1	Loose	81.344	5.78%	1.571
SP_Johnson_8	Aggressive	209.828	0.31%	0.069
SP_Johnson_8	Baseline	212.974	-1.19%	0.367
SP_Johnson_8	Loose	210.690	-0.10%	1.473
SP_Winnebago_1	Aggressive	87.666	2.71%	0.104
SP_Winnebago_1	Baseline	88.185	2.13%	0.382
SP_Winnebago_1	Loose	84.112	6.65%	2.037

VII. ROBUSTNESS STUDY

The core results suggest that screened rollout can improve upon deterministic mean-time replanning, but they also raise a natural question: are these gains robust to different modeling and algorithmic choices? This question matters because the original SP_* instances are deterministic, while this study adds a synthetic stochastic execution layer on top of the benchmark data. The robustness study therefore examines three dimensions: action-screening intensity, uncertainty specification, and induced-subinstance seed.

A. Robustness Design

All robustness experiments use induced 20-customer instances and compare the same three policies as before: open-loop mean planning, deterministic mean-time replanning, and screened rollout. The main performance metric is the percentage improvement of rollout relative to mean-time replanning in mean total completion time. Within each robustness setting, the policies are evaluated on a common set of episode seeds.

B. Action-Screening Robustness

This check varies how aggressively the offline sortie library is filtered before rollout evaluation. Aggressive screening produces a smaller candidate-action set, the baseline setting matches the core experiment matrix, and loose screening allows a larger set of line and loop actions to survive to the rollout stage.

Table V shows that candidate-action richness matters for the urban and mixed instances. On SP_Cook_1, rollout improves over mean replanning under all three screening rules, but the improvement increases from 1.41% under aggressive screening to 5.78% under loose screening. A similar pattern appears on SP_Winnebago_1, where loose screening gives the largest improvement of 6.65%. These results indicate that restrictive screening can remove useful coordination actions before the Monte Carlo evaluation stage.

The benefit of looser screening comes with a computational cost. On SP_Cook_1, the average rollout decision time increases from 0.079 seconds under aggressive screening to 1.571 seconds under loose screening. On SP_Winnebago_1, it increases from 0.104 seconds to 2.037 seconds. Thus, screening induces a clear quality-runtime tradeoff. A richer

TABLE VI
UNCERTAINTY-SPECIFICATION ROBUSTNESS ON INDUCED 20-CUSTOMER
INSTANCES.

Instance	Specification	Rollout	Imp. vs Replan	Time (s)
SP_Cook_1	Baseline	83.057	3.80%	0.470
SP_Cook_1	I.i.d. regime	83.804	2.69%	0.715
SP_Cook_1	Service-dominant	82.730	5.33%	0.576
SP_Cook_1	Truck-dominant	83.332	3.13%	0.587
SP_Cook_1	Uniform noise	83.076	3.78%	0.448
SP_Johnson_8	Baseline	212.974	-1.19%	0.363
SP_Johnson_8	I.i.d. regime	207.369	0.79%	0.334
SP_Johnson_8	Service-dominant	206.536	-0.88%	0.337
SP_Johnson_8	Truck-dominant	210.556	5.84%	0.307
SP_Johnson_8	Uniform noise	211.183	-0.62%	0.374
SP_Winnebago_1	Baseline	88.185	2.13%	0.379
SP_Winnebago_1	I.i.d. regime	90.496	-0.02%	0.728
SP_Winnebago_1	Service-dominant	91.501	-0.18%	0.581
SP_Winnebago_1	Truck-dominant	91.927	-3.15%	0.398
SP_Winnebago_1	Uniform noise	88.425	1.75%	0.386

candidate set can improve decision quality, but it also makes online decision making slower.

The rural instance SP_Johnson_8 behaves differently. Rollout is close to mean replanning under all three screening rules. This pattern suggests that the difficulty of the rural instance is not fully explained by overly restrictive screening. Instead, the current sortie library or base policy may be less well matched to this geographic structure.

C. Uncertainty-Specification Robustness

This check changes the source and structure of uncertainty rather than only its overall level. The baseline medium-uncertainty model is compared with four alternatives: an i.i.d. traffic-regime model with weaker temporal persistence, a service-dominant model, a truck-dominant model, and a uniform truck-noise model that removes the drive-walk-based heteroskedasticity.

Table VI shows that the urban instance SP_Cook_1 is robust across all uncertainty specifications. Rollout improves over mean replanning in every case, with gains ranging from 2.69% under the i.i.d. regime model to 5.33% under the service-dominant model. This indicates that the rollout advantage on the urban instance is not tied to one particular stochastic model.

The rural instance SP_Johnson_8 is more sensitive to the source of uncertainty. Under the baseline and service-dominant specifications, rollout slightly underperforms mean replanning. Under truck-dominant uncertainty, however, rollout improves by 5.84%. This result is consistent with the synchronization logic of the truck-walk system. When truck travel times are more uncertain, online stochastic lookahead can better account for future waiting and reunion risk.

SP_Winnebago_1 shows the least stable pattern across uncertainty specifications. Rollout improves under the baseline and uniform-noise settings, is nearly tied under i.i.d. regime and service-dominant uncertainty, and underperforms under truck-dominant uncertainty. This indicates that the mixed instance is more sensitive to how uncertainty enters the execution process.

TABLE VII
SUBINSTANCE-SEED ROBUSTNESS ON INDUCED 20-CUSTOMER
INSTANCES.

Instance	Seed	Rollout	Imp. vs Replan	Time (s)
SP_Cook_1	1	81.843	6.16%	0.442
SP_Cook_1	2	87.458	4.53%	0.545
SP_Cook_1	3	85.775	4.21%	0.760
SP_Cook_1	4	89.188	5.26%	0.285
SP_Cook_1	5	90.562	0.61%	0.532
SP_Johnson_8	1	205.103	6.19%	0.358
SP_Johnson_8	2	175.027	0.86%	0.742
SP_Johnson_8	3	210.443	4.34%	0.498
SP_Johnson_8	4	214.399	7.88%	0.317
SP_Johnson_8	5	162.234	5.72%	0.245
SP_Winnebago_1	1	85.750	-0.36%	0.401
SP_Winnebago_1	2	90.354	4.00%	0.777
SP_Winnebago_1	3	84.320	9.71%	0.248
SP_Winnebago_1	4	83.444	5.63%	0.299
SP_Winnebago_1	5	89.714	0.06%	0.555

D. Subinstance-Seed Robustness

This check keeps the customer count fixed at 20 but varies the induced-subinstance seed, so each row corresponds to a different sampled customer subset from the original 50-customer benchmark instance. The purpose is to determine whether the rollout improvement is driven by one favorable induced instance.

Table VII shows that rollout performance is generally stable across sampled subinstances. On SP_Cook_1, rollout improves over mean replanning for all five seeds, with gains ranging from 0.61% to 6.16%. This reinforces the earlier finding that the urban instance provides the most reliable setting for screened rollout.

The results on SP_Johnson_8 are also informative. Although the main experiment under the baseline seed showed a slight negative result for the 20-customer rural instance, rollout improves over mean replanning for all five alternative seeds here. This indicates that the earlier rural weakness is not a universal failure of rollout, but is sensitive to the particular customer subset sampled from the benchmark instance.

SP_Winnebago_1 shows more variation. Rollout improves in most seeds, with the largest gain reaching 9.71%, but it slightly underperforms under seed 1 and is nearly tied under seed 5. This indicates that the mixed instance is more sensitive to customer composition than the urban instance.

E. Remarks

Overall, the robustness study leads to three main conclusions. First, action screening is a key design choice rather than only a computational shortcut. Looser screening can improve rollout performance on urban and mixed instances, but it also increases online decision time. Second, the value of rollout depends on the structure of uncertainty, not only its magnitude. In particular, rollout is robustly beneficial on the urban instance and becomes especially valuable on the rural instance when truck-side uncertainty dominates. Third, the subinstance-seed results suggest that the observed gains

are not driven by a single favorable customer subset, although the size of the improvement varies across sampled instances.

F. Limitations

Several limitations should be noted. First, the uncertainty model is synthetic, although it is informed by benchmark structure. The paper does not estimate travel-time or service-time distributions from historical operations data. Second, the action space is intentionally restricted through an offline sortie library and online screening. This restriction is necessary for tractability, but it also limits the policy class. Third, the full-size experiments were conducted on selected instances rather than on the entire benchmark set. These points do not invalidate the findings, but they do define the scope of the conclusions.

VIII. CONCLUSION

This paper formulates truck-walk execution under uncertainty as a finite-horizon sync-point MDP and solves it approximately with a screened staged rollout policy. The computational study shows that this approach can improve upon deterministic baselines on representative SP_* instances while remaining computationally tractable. The strongest gains appear on urban and mixed instances, and the full-size 50-customer runs show that the method can scale beyond very small instances. The main remaining limitation is robustness on some induced rural instances, where additional action-design or base-policy improvements may still be needed.

Future work could incorporate richer waiting penalties or develop stronger post-decision approximations for large rural instances.

IX. CONTRIBUTIONS AND RELEASE

The original idea and proposal were developed by Chenhui Qu. In the final stage of the study, Chenhui Qu implemented the algorithms, conducted and analyzed the experiments, and wrote the report. Ruogu Sun added the robustness study. Chenhui Qu and Ruogu Sun jointly revised the final materials.

All algorithms used in this study were implemented from scratch in Python using the standard library. This includes the sync-point MDP simulator, the regime-aware uncertainty model, the offline action library, the deterministic replanning baseline, the staged rollout policy, and the experiment scripts. No off-the-shelf routing or MDP solver was used.

The authors grant permission for this report to be posted publicly.

REFERENCES

- [1] S. Reed, A. M. Campbell, and B. W. Thomas, "The Value of Autonomous Vehicles for Last-Mile Deliveries in Urban Environments," *Management Science*, vol. 68, no. 1, pp. 280–299, 2022, doi: 10.1287/mnsc.2020.3917.
- [2] S. Reed, A. M. Campbell, and B. W. Thomas, "Impact of Autonomous Vehicle Assisted Last-Mile Delivery in Urban to Rural Settings," *Transportation Science*, vol. 56, no. 6, pp. 1530–1548, 2022, doi: 10.1287/trsc.2022.1142.
- [3] R. Zhang, "Branch-and-Price for the Capacitated Autonomous Vehicle Assisted Delivery Problem," *INFORMS Journal on Computing*, vol. 37, no. 5, pp. 1328–1349, 2025, doi: 10.1287/ijoc.2023.0177.

- [4] N. Secomandi, "A Rollout Policy for the Vehicle Routing Problem with Stochastic Demands," *Operations Research*, vol. 49, no. 5, pp. 796–802, 2001, doi: 10.1287/opre.49.5.796.10608.
- [5] J. C. Goodson, J. W. Ohlmann, and B. W. Thomas, "Rollout Policies for Dynamic Solutions to the Multivehicle Routing Problem with Stochastic Demand and Duration Limits," *Operations Research*, vol. 61, no. 1, pp. 138–154, 2013, doi: 10.1287/opre.1120.1127.
- [6] J. C. Goodson, B. W. Thomas, and J. W. Ohlmann, "Restocking-Based Rollout Policies for the Vehicle Routing Problem with Stochastic Demand and Duration Limits," *Transportation Science*, vol. 50, no. 2, pp. 591–607, 2016, doi: 10.1287/trsc.2015.0591.
- [7] M. W. Ulmer, J. C. Goodson, D. C. Mattfeld, and M. Hennig, "Offline–Online Approximate Dynamic Programming for Dynamic Vehicle Routing with Stochastic Requests," *Transportation Science*, vol. 53, no. 1, pp. 185–202, 2019, doi: 10.1287/trsc.2017.0767.
- [8] S. Zhang, J. W. Ohlmann, and B. W. Thomas, "Dynamic Orienteering on a Network of Queues," *Transportation Science*, vol. 52, no. 3, pp. 691–706, 2018, doi: 10.1287/trsc.2017.0761.